

Sri Mallika Ponnada
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AI/ ML Engineer



Professional Summary:

- Over 4 years of experience in Data Analytics, Machine Learning (ML), Natural Language Processing (NLP), and Generative AI, delivering end-to-end solutions across Finance, Telecom, and Legal domains.
- Hands-on experience across the end-to-end data science lifecycle: data acquisition, cleaning, feature engineering, model development, evaluation, and deployment on cloud platforms like AWS and Azure.
- Developed ML models using Random Forest, XGBoost, SVM, Decision Trees, Naive Bayes, and Deep Neural Networks, applied in real-world use cases such as fraud detection, customer segmentation, and intelligent document processing.
- Strong foundation in NLP techniques including tokenization, stemming, lemmatization, POS tagging, and Named Entity Recognition using spaCy, Hugging Face Transformers, and LangChain.
- Implemented Generative AI and LLM-based applications using GPT, Claude, and Bedrock, including Retrieval-Augmented Generation (RAG) pipelines and intelligent chatbots for knowledge discovery and semantic search.
- Experienced in building enterprise-grade Agentic RAG architectures and integrating advanced LLM capabilities such as GPT-5.1 reasoning and Model Context Protocol (MCP) for secure, context-grounded AI applications.
- Developed and optimized data pipelines using Python, SQL, Spark, and Scala, and built efficient ETL workflows on Azure Data Factory, AWS Lambda, and SageMaker for processing structured and unstructured data.
- Proficient in data visualization and storytelling using Power BI, Tableau, and Spotfire, enabling actionable insights and reporting for diverse stakeholders across business units.
- Experience working with various databases such as MySQL, PostgreSQL, MongoDB, Amazon Redshift, and Azure Synapse Analytics, with expertise in SQL query optimization and data retrieval.
- Familiar with CI/CD and containerized deployment practices using Docker, Kubernetes, Jenkins, and Terraform, ensuring smooth integration and scalability of ML services.
- Strong understanding of statistical analysis, A/B testing, hypothesis testing, and time-series analysis for developing robust predictive models.
- Adept at collaborating with cross-functional teams to translate business problems into ML solutions and deliver data-driven outcomes in production environments.
- Delivered scalable AI/ML solutions using cloud-native services including Amazon Bedrock, OpenSearch, CloudWatch, and Azure Synapse, reducing latency and improving system observability.

Technical Skills

Technologies & Tools:	Machine Learning (Random Forest, XGBoost, SVM, Decision Trees), Deep Learning (LSTM, Transformers, GANs), Generative AI (GPT, Claude, LLaMA), Natural Language Processing (spaCy, Hugging Face Transformers, LangChain), Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone, FAISS), Data Analytics, Data Mining, Data Engineering, Statistical Analysis, Predictive Modeling, Power BI, Tableau, Spotfire, Apache Spark, ETL (Azure Data Factory, AWS Glue), CI/CD (Jenkins, Bitbucket, MLFlow)
Languages:	Python, SQL,.NET
Cloud & Deployment	AWS (Lambda, SageMaker, Bedrock, API Gateway, Redshift, OpenSearch, CloudWatch), Azure (Synapse, Functions), GCP (basic familiarity), Docker, Kubernetes, Terraform, AI Foundry

Databases:

MySQL, PostgreSQL, MongoDB, Amazon Redshift, Azure Synapse Analytics, cosmos DB

PROFESSIONAL EXPERIENCE:

Client: AMEX, New York, United States

Jan 2024 – Till date

Role: Gen AI/ ML Engineer

Project: AI-Powered Fraud Detection and Customer Support Chatbot for Enhanced Security and Service Efficiency

Description: Played a key role in developing and deploying an AI-driven customer support chatbot integrated with a real-time fraud detection system. The solution improved automated customer interactions, accelerated fraud resolution, and increased the accuracy of responses for transaction inquiries and security concerns. Leveraged Generative AI models and LLMs like Claude and GPT, utilizing retrieval-augmented generation (RAG) techniques to provide context-aware, accurate responses grounded in AMEX proprietary data.

Functional role details:

- Developed a RAG-based framework to vectorize and semantically index historical customer interactions, transaction sequences, and fraud alert data, enabling efficient retrieval of relevant information for both chatbot responses and fraud detection.
- Utilized GPT-5.1's enhanced reasoning and structured output capabilities to strengthen fraud-analysis workflows and deliver more accurate, context-aligned responses within the customer support chatbot.
- Used Amazon OpenSearch Service to store, index, and query vectorized customer conversations and transaction records for rapid retrieval during chatbot interactions and fraud investigations.
- Implemented Amazon Lambda to invoke LLMs such as GPT and Claude in real time, providing scalable, cost-efficient AI-driven responses within the chatbot and fraud detection workflows.
- Implemented intelligent response-caching and dynamic routing strategies to reduce redundant LLM calls, optimizing GPT and Claude usage for lower inference cost and improved latency.
- Leveraged AWS Bedrock to provision, manage, and scale LLM models, supporting growing volumes of queries related to customer service and fraud alerts without compromising response accuracy or latency.
- Employed vector databases like Pinecone and FAISS (via OpenSearch) to optimize similarity search across large-scale transactional and conversational embeddings, ensuring contextually relevant and trustworthy responses.
- Built an Agentic RAG architecture enabling coordinated retriever, ranker, verifier, and explainer agents to improve evidence validation, reduce hallucinations, and enhance decision reliability across fraud-related workflows.
- Integrated Model Context Protocol (MCP) to enable secure interoperability between LLMs and AMEX backend systems, ensuring real-time grounding of customer and transaction data for consistent, compliant AI reasoning.
- Designed reusable prompt templates and parameterized prompt strategies for fraud explanation, transaction summarization, and chatbot escalations, improving LLM response consistency and reducing latency across workflows.
- Integrated Amazon CloudWatch, Prometheus, and Grafana to monitor system health, fraud model accuracy, chatbot latency, and overall AI performance in real time, ensuring stable and reliable operations.
- Established CI/CD pipelines with AWS CodePipeline and Jenkins to automate retraining and deployment of fraud detection models and chatbot updates, adapting swiftly to emerging fraud patterns and customer service changes.
- Enabled customers to ask detailed questions such as "Why was this transaction flagged as suspicious?" or "Can you explain the recent charges on my account?" and receive precise, AI-generated answers, reducing live agent workload by 30% and improving fraud resolution times.

Client: Thomson Reuters corporation, India

May 2021 – June 2022

Role: Data scientist/ ML Engineer

Project: AI-Based Career Path & Learning Recommendation Systems

Description: Contributed to the development of a machine learning system that recommended personalized career paths and learning content for internal users. The solution supported employee development and improved engagement through AI-powered recommendations.

Functional role details:

- Built and deployed ML models using Python to recommend personalized learning paths within the company's LMS platform.
- Integrated predictive analytics into internal career sites to suggest growth opportunities aligned with employee skills and interests.
- Utilized ML algorithms such as decision trees, graph-based models, and classification techniques to improve recommendation relevance and model interpretability.
- Developed normalized and denormalized data models to enhance database query performance by up to 40%.
- Collaborated with cross-functional teams to define data flows and validate model integration within business applications.
- Conducted end-to-end testing of ML model pipelines to ensure accuracy, stability, and seamless application performance.
- Engineered features from user behavior, skill profiles, and historical learning patterns to enhance model accuracy and personalization.
- Processed and analyzed real-time user interaction data to continuously refine recommendation algorithms and improve accuracy.
- Developed SQL-based ETL pipelines to prepare structured datasets from employee activity logs and LMS records, ensuring clean and reliable data for ML workflows.
- Contributed to increased employee engagement by improving the relevance and personalization of AI-driven career and learning recommendations.

Client: Blue I Technologies Pvt Ltd, India

May 2020 – April 2021

Role: Data Analyst

Project: Customer Segmentation & Sales Insights for Retail Client

Contributed to a retail analytics initiative focused on enhancing customer understanding, campaign effectiveness, and sales growth through data-driven decision-making.

- Performed data preparation and exploratory analysis for a retail client using SQL and Python (pandas).
- Created analytical reports and visualizations in Python and Power BI, presenting customer purchase patterns and business-critical metrics.
- Collaborated with business and marketing teams to translate analytical findings into actionable insights for campaign planning and customer targeting.
- Enhanced an ML-based customer segmentation model by categorizing customers into purchasing-power groups, enabling targeted campaigns that contributed to over \$1M in sales in 2020.
- Conducted A/B testing to evaluate marketing campaign effectiveness and identify high-performing strategies
- Built automation scripts using SQL and Python to streamline recurring reporting workflows and reduce manual effort.

Certifications:

AWS Certified Solution Architect

Azure AI Fundamentals (AI-900)

Databricks Generative AI Certified

Education:

Master of Information Systems

Cleveland State University, Monte Ahuja College of Business, Cleveland, OH